

Mars, warm and wet?



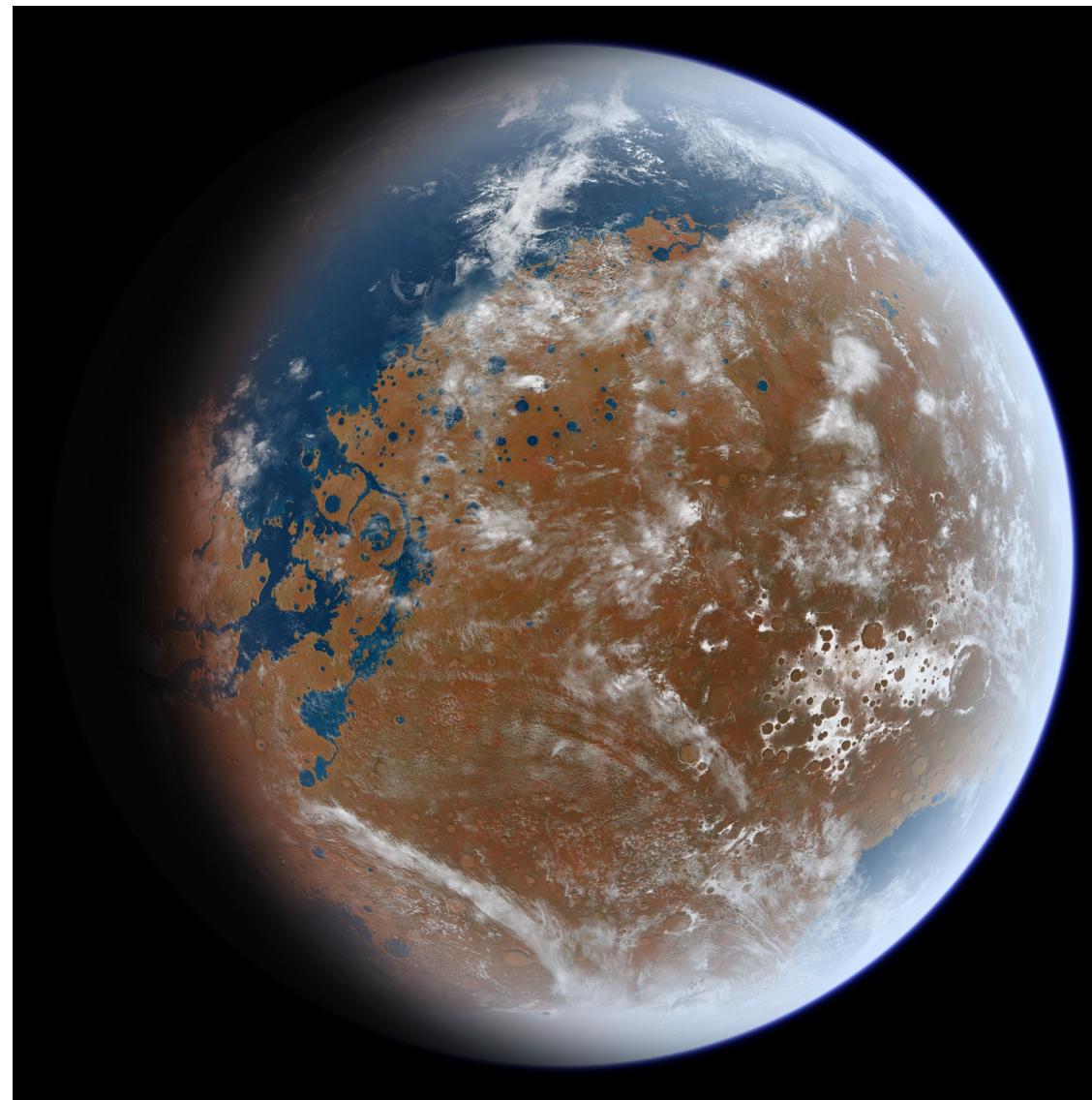
An artistic depiction of a warm and wet Mars (left) believed to have a dense atmosphere and water, and a present dry and cold Mars (right)

Image Credit: NASA's Goddard Space Flight Center



With many different missions and long observations of the planet Mars, enough data has been gathered to theorise that long ago the Martian surface was warm and wet. With many channels carved into its dry surface, the analysis of the composition of its surface and current atmosphere and changes in it, via rovers and satellite shows that once Mars was covered in water and had a warm dense atmosphere.

Mars has been the centre of Space exploration focus for the past few decades, with the goals of exploring not only the planet itself but its evolution throughout its history. As humans started to explore the Red Planet in search of biosignatures, it was essential to learn about its past environment and how it was different compared to present day Mars. Whether Mars ever supported life or not is still a big mystery, but the scientists know that the answer to this mystery lies within understanding and clearly mapping out the evolution of Martian environment. Mariner 9, Viking 1 and 2 were the first step towards mapping the Martian environment and collecting topological data that would help the scientists to start piecing together the mysteries of our neighbouring planet. With the help of these missions and their results, the peak in the exploration of Mars grew with many more missions in the following decades. These missions were dedicated towards exploring Martian surface, analysing its atmosphere and surface and start searching for signs of life.



Artistic depiction of what Oceans and sea look like during the Paleo-Ocean and Ocean Borealis based on the geological data.

Evidence of a different environment

The Martian surface offers much evidence and insight into its thriving past as a warm planet covered in water with dense atmosphere. The evidence can be categorised into geological and atmospheric data for better understanding.

Quick Summary of Martian Geological History

Pre Noachian Era (from 4-5Gyr to 4.1Gyr)

From the formation of planet to formation of Hella

Noachian Era (from 4.1Gyr to 3.7Gyr ago)

warm atmosphere, formation of valley networks, presence of water, volcanic activities, formation of many phyllosilicates and high erosion rates of crater during the time period.

Hesperian Era (from 3.7Gyr to 3Gyr)

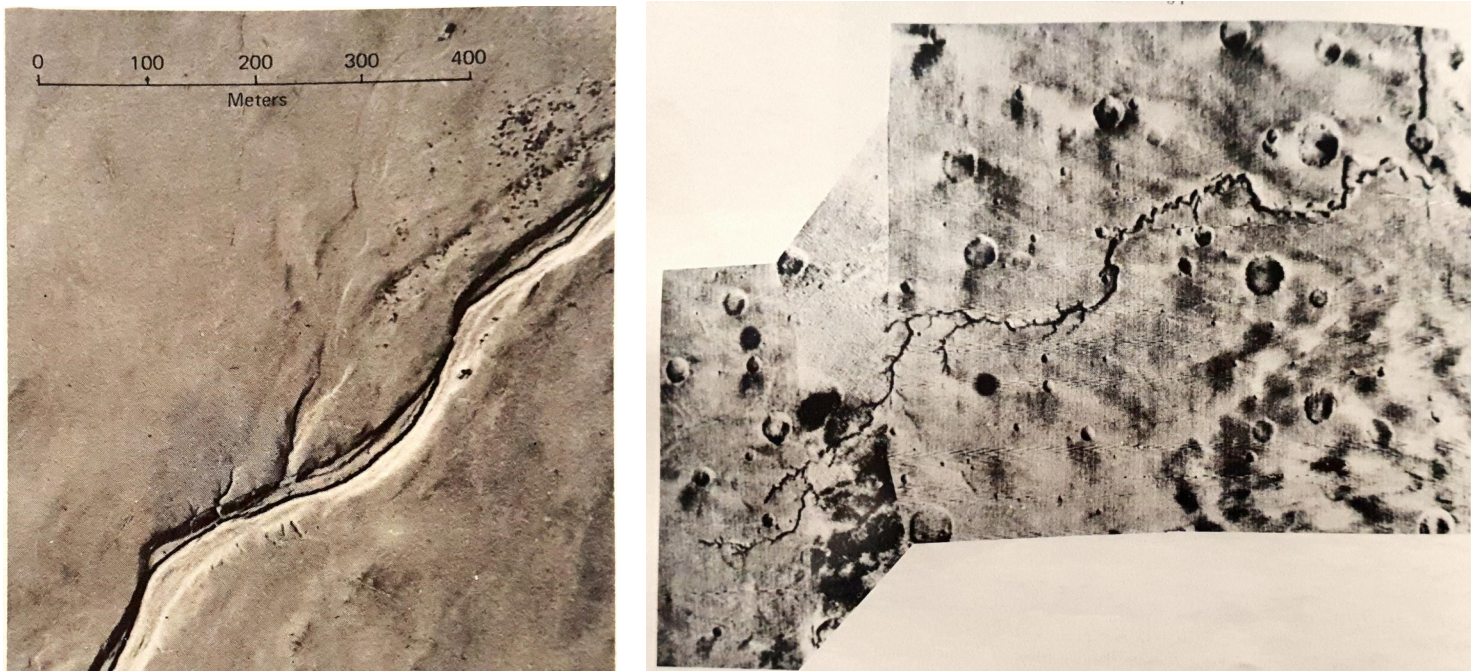
Formation of lava plains, canyons, comparatively low rates of valley formation, formation of largest outflows and terminal lakes or seas, cease or very low rates of rock alteration to form phyllosilicates, sulphate rich deposits and very low rates of erosions. Rare presence of favourable atmosphere for liquid water, possibly lost of thick atmosphere

Amazonian Era (from 3.7Gyr to present day)

Very small changes, decline in erosion rates continued, changes in the surface by tectonism and crater impact is modest compared to earlier time periods.

>> Geological Features

The geological features of Mars convey its mystery wrapped history to the new age explorer, us humans in a cipher. When these features, first discovered by the Mariner 9 were compared to their terrestrial counterparts they portrayed riverbeds and shore lines pointing towards presence of river and channel networks along with oceans, seas and lake on the Red planet, pointing towards an atmosphere stable enough to support the presence of liquid water for a substantial period.

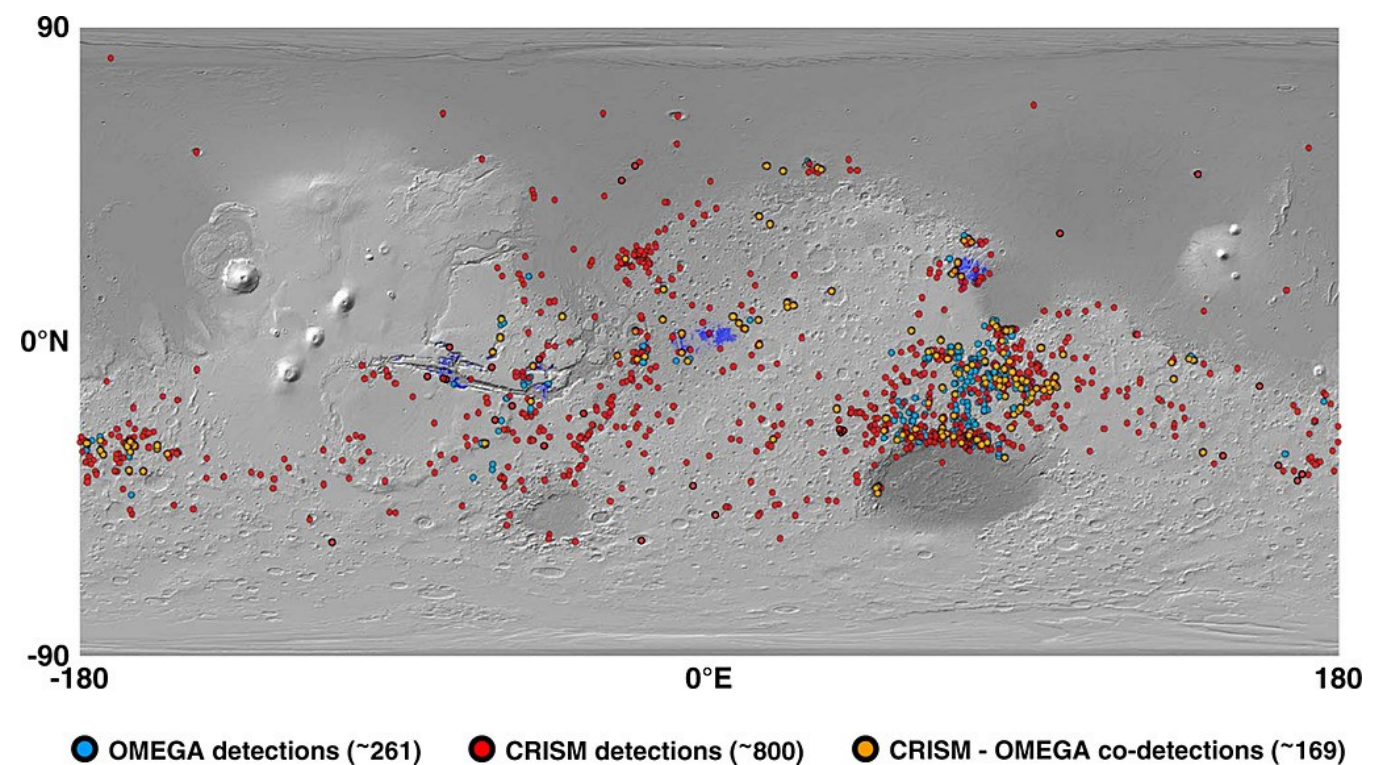


Channel cut in an extremely dry region in the Coastal desert of Peru. (left. Terrestrial example) Mosaic of Vallis Nirgal in Mare Erythraeum showing details of the tributary system and central meandering portion. Length of the channel is 1200km. (right, Martian channel)

Image Credit: U.S geological Survey (left), JPL (right)

The claims are not only supported through visible similarities between the geological features, surface analysis show presence of alluvial deposits and clays in such regions which are predicted to be part of the planet's fluvial system. Data collected from Mars Orbiter Laser Altimeter (MOLA), CRISM (Compact Reconnaissance Imaging Spectrometer for Mars) and OMEGA (Observatoire pour la Minéralogie, l'Eau, les Glaces et l'Activité/

Visible and Infrared Mineralogical Mapping Spectrometer) shows the presence of hydrous minerals across the slopes of craters and channels. The results from the orbiters along with the data collected from rovers show presence of phyllosilicate clays, Fe/Mg smectites, carbonates and iron hydroxides/oxyhydroxides. These minerals suggest presence of water during alteration of surface minerals, as some of these require long term aqueous activity while some require short term hydrothermal events and little moisture.



Martian map showing the detection of hydrous minerals by OMEGA and CRISM.

>> Summary of Geological Features

Valley Networks:

In the late 70s the mariner 9 discovered valleys very similar to river drainage valleys on the planet Earth. Small channels combine into larger valleys which would lead to the formation of dendritic networks. The presence of these valley networks is only observed on surfaces of the Noachian era (older than 3.8-3.5 Gyr), signaling a change in climate on Mars which affected the presence of water on the planet. Although the method of the formation of these networks is still under speculation it is widely accepted that liquid water was involved in the formation of these networks over a substantial time.

Lacustrine Deposits:

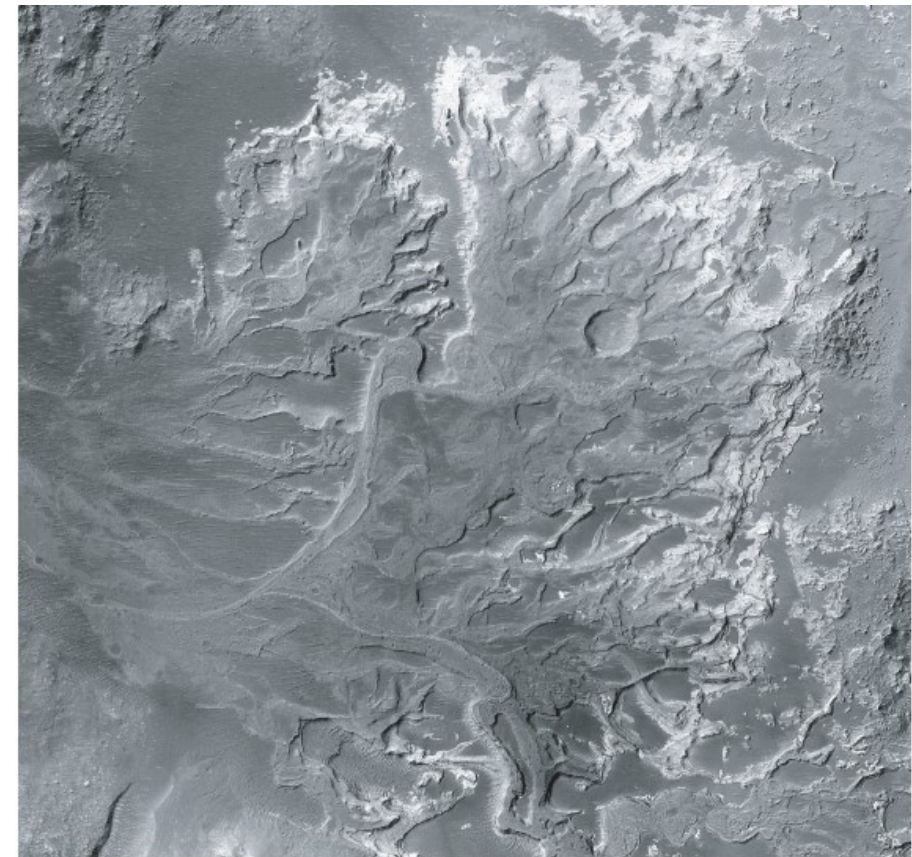
Images and data from the Mars Surveyor Camera first discovered the evidence of deposits inside the crater resembling layered lacustrine deposits, deltas, and shorelines. It suggested these deposits were buried for a long time and have only surfaced recently. Studies for ExoMars landing site Oxia Planum have shown this to be the case indeed. Lacustrine deposits predict the presence of standing bodies of water on Mars in the far past.

Erosion Rates:

The craters from the Noachian era show high levels of erosion compared to craters from later periods. Craters of diameters of 15km have very degraded appearances and those smaller than 15km have been completely erased, whereas craters only a few hundred million years younger are well preserved, pointing towards a much higher and more efficient erosion rate during that period. Although many possible causes can affect the erosion rate, it is thought that liquid water was the eroding agent.

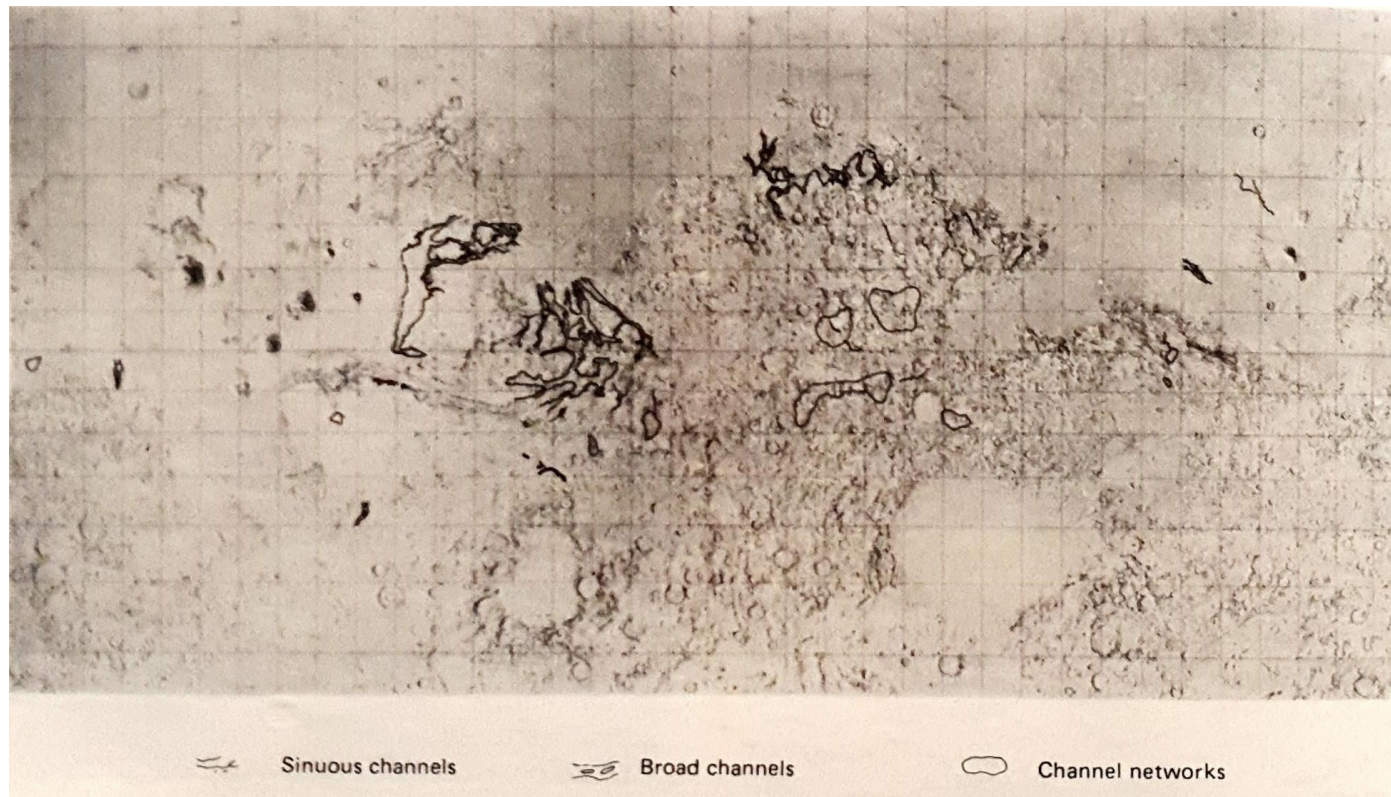
Minerology :

Data from rovers and orbits such as TES, OMEGA, and CRISM, show the presence of hydrous sulfates, clays, and mineral hematite. Whereas the sulfate-rich deposits and the hematite are present in locations and terrains from both Hesperian and Noachian eras, phyllosilicates are only present in locations from the Noachian era. The process required for rock alteration to form phyllosilicates suggests that there were fluvial activities on the planet for a few hundred million years.



Remnants of sedimentary alluvial fans analogous to a river delta. These ancient landmark characteristics, which probably have been covered by layers of materials which then eroded allowing these sediments to resurface allow us to predict that some valleys on Mars experience persistent, on going flow of water over long periods of times, as rivers do on Earth.

Image Credit: MOC camera, Mars Global Surveyor, NASA/JPL/MSSS



Types of Channel formation mapped by the mariner 9 on Martian Surface.

Image Credit: NASA

>> Evidence of an Atmosphere

Observing atmospheric data alongside geological data on Mars, the scientist has reached the conclusion that Mars once had a warm and thick atmosphere which was later lost to space. Mars Atmosphere and Volatile Evolution, MAVEN was launched by NASA in the November of 2013 to study the atmospheric changes in Mars atmosphere to determine how much atmosphere has Mars lost to space. The studies would help researchers analyse

and predict for how long Mars was a habitable planet and when the changes in the atmosphere made it hostile to life. Based on the analytics of the data from MAVEN, the NASA scientist predicts that Mars have lost 2/3 of its atmosphere to space from solar winds after it lost its magnetic fields. One observation that points towards a dense greenhouse filled atmosphere on Mars, is the evidence of the presence of liquid water during the period when the luminosity of the Sun was only 75% of the current luminosity, meaning the surface temperature of the planet would have -75°C without accounting for the atmosphere. The only possible way for water to exist in liquid form during that era was only if there was a dense atmosphere of greenhouse gases on the planet.

Martian Surface at a Glance

- > Dry surface, red due to presence of iron oxide
- > Very thin atmosphere composed of mostly CO_2 (95.3%) N (2.7%) and Ar (1.6%)
- > Atmospheric pressure 610 Pa
- > Surface temperature average -53°C
- > Varies from -123°C to 27°C depending on the latitude and time of the year.

Did Mars had a warm and wet environment ?

Since the first mission to Mars, and after many missions and joint efforts of NASA, ESA, JAXA, and other space companies we have discovered and collected data and evidence that allow us to predict a Mars where water was available in a liquid state in form of rivers, lakes, and ocean. But all these scenarios are predictions and guesses and the presence of water is one of the few possible scenarios. Based on the analysis of the gathered data, Mars seemed to have a warm climate during the Noachian era (from 4.1 Gyr to 3.7 Gyr years ago) which allowed the presence of liquid water on the planet for a few hundred million years, surfaces from this period show have landmark characteristics that support the theory of alluvial activities during the era. Although with the change in climate, much-predicted loss of atmosphere to the solar winds the planet's environment was not suitable for liquid water, which over a few million years changed the planet to what we know it today.

>>Quick Links

- > <https://mars.nasa.gov/maven/>
- > https://www.esa.int/Science_Exploration/Space_Science/Mars_Express
- > https://www.esa.int/Science_Exploration/Human_and_Robotic_Exploration/Exploration/ExoMars
- > https://mars.nasa.gov/mars-exploration/missions/?page=0&per_page=99&order=date+desc&search=

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- Martian Stable Isotopes: Volatile Evolution, Climate Change and Exobiological Implications. *Orig Life Evol Biosph* **29**, 47–57 (1999). <https://doi.org/10.1023/A:1006586319129> Jakosky, B.M.
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- Hydrous minerals on Mars as seen by the CRISM and OMEGA imaging spectrometers: Updated global view, [J. Carter](#), [F. Poulet](#), [J.-P. Bibring](#), [N. Mangold](#), [S. Murchie](#) ,
- Donald R. Lowe, Janice L. Bishop, Damien Loizeau, James J. Wray, Ross A. Beyer; Deposition of >3.7 Ga clay-rich strata of the Mawrth Vallis Group, Mars, in lacustrine, alluvial, and aeolian environments. *GSA Bulletin* 2019;; 132 (1-2): 17–30. doi: <https://doi.org/10.1130/B35185.1>